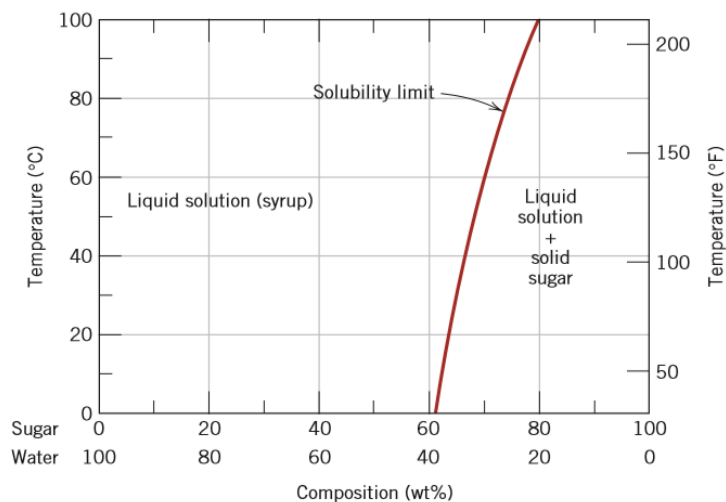
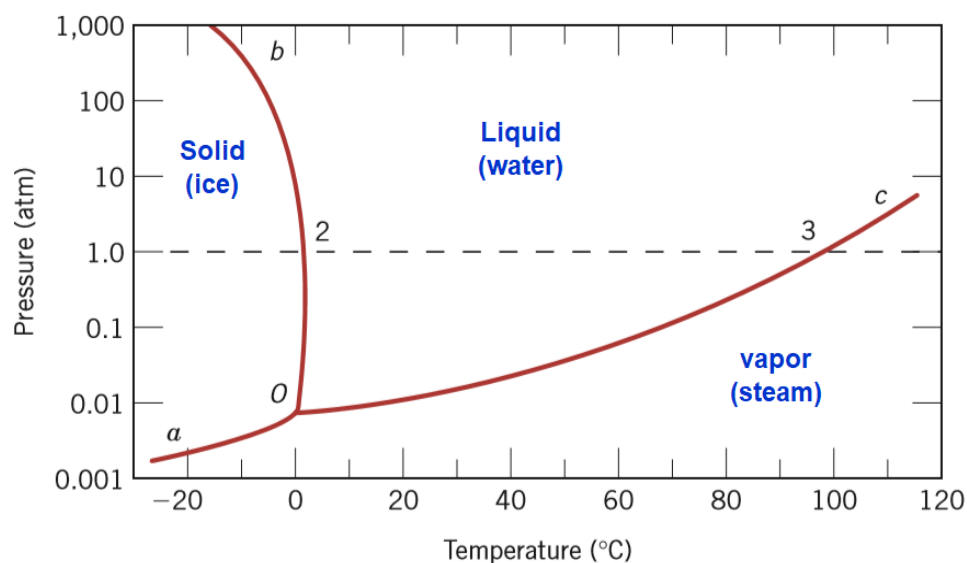


WS N°2: Phase Diagrams (Part I)

1. Consider the sugar–water phase diagram in the figure below.
 - a. How much sugar will dissolve in 1500 g water at 90 °C (194 ° F) ?
 - b. If the saturated liquid solution in part (a) is cooled to 20 °C (68 ° F), some of the sugar will precipitate out as a solid. What will be the composition of the saturated liquid solution at 20 °C ?
 - c. How much of the solid sugar will come out of solution upon cooling to 20 °C ?



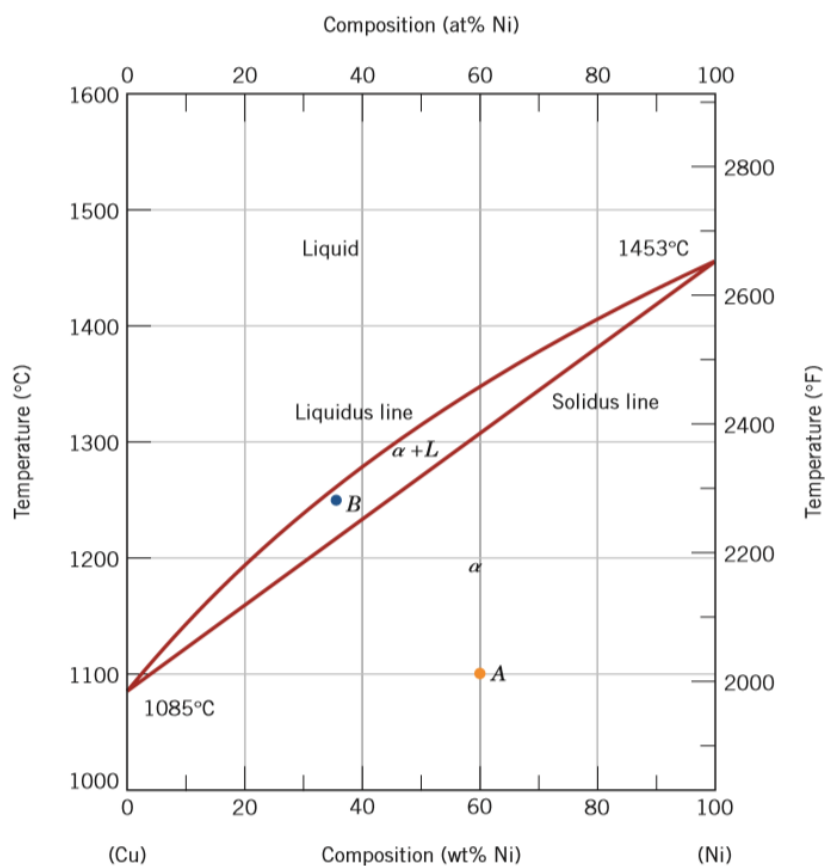
2. Consider a specimen of ice that is at -10 °C and 1 atm pressure. Using the figure of the pressure– temperature phase diagram for H₂O below, determine the pressure to which the specimen must be raised or lowered to cause it
 - a. to melt
 - b. to sublime.



3. Using the figure in the previous exercise (Number 2) at a pressure of 0.01 atm, determine
 - a. the melting temperature for ice
 - b. the boiling temperature for water.
4. In the table below, the solidus and liquidus temperatures for the germanium-silicon system are given. Construct the phase diagram for this system and label each region.

Composition (wt% Si)	Solidus Temperature (°C)	Liquidus Temperature (°C)
0	938	938
10	1005	1147
20	1065	1226
30	1123	1278
40	1178	1315
50	1232	1346
60	1282	1367
70	1326	1385
80	1359	1397
90	1390	1408
100	1414	1414

5. Using the phase diagram below
 - a. Determine the phase(s) present for the Nickel-Copper alloy having 8.2 moles of nickel and 4.3 moles of copper at 1250°C (2280°F). Knowing that the molar mass of nickel is 58.6934 amu and that of copper is 63.5464 amu.
 - b. Compute the relative amount in terms of mass fraction of the phase(s) present for the Nickel-Copper alloy at 1250°C.



6. Is it possible to have a copper–nickel alloy that, at equilibrium, consists of a liquid phase of composition 20 wt% Ni–80 wt% Cu and also an α phase of composition 37 wt% Ni–63 wt% Cu? If so, what will be the approximate temperature of the alloy? If this is not possible, explain why.
7. Is it possible to have a copper–zinc alloy that, at equilibrium, consists of an ϵ phase of composition 80 wt% Zn–20 wt% Cu, and also a liquid phase of composition 95 wt% Zn–5 wt% Cu? If so, what will be the approximate temperature of the alloy? If this is not possible, explain why.

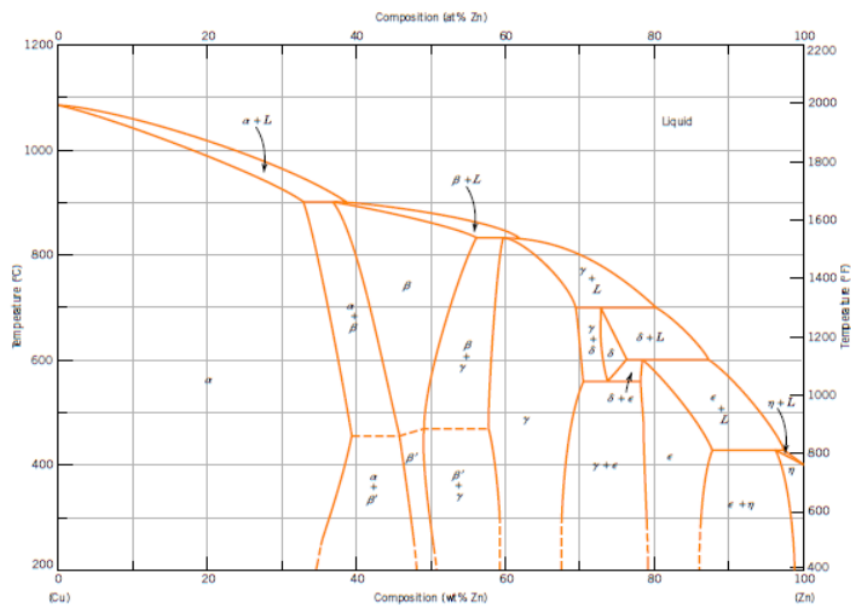


Figure 9.19 The copper–zinc phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 2, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

8. 1.5-kg specimen of a 90 wt% Pb–10 wt% Sn alloy is heated to 250°C (480°F); at this temperature it is entirely a phase solid solution (Figure 9.8). The alloy is to be melted to the extent that 50% of the specimen is liquid, the remainder being the phase. This may be accomplished either by heating the alloy or changing its composition while holding the temperature constant.
 - a. To what temperature must the specimen be heated?
 - b. How much tin must be added to the 1.5-kg?

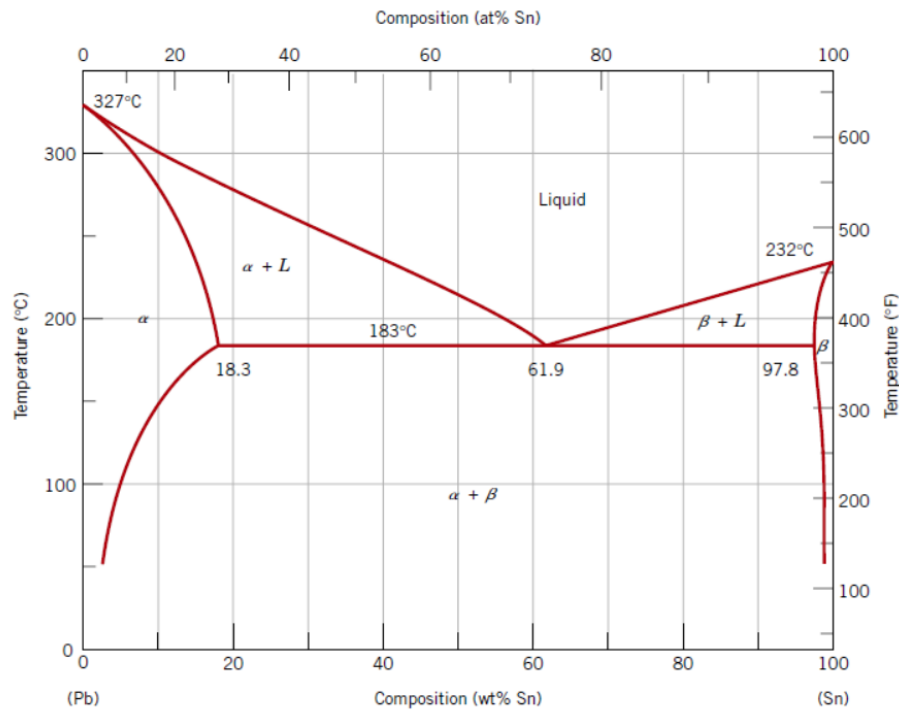


Figure 9.8 The lead–tin phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 3, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

9. A magnesium–lead alloy of mass 5.5 kg consists of a solid α phase that has a composition just slightly below the solubility limit at 200°C (390°F).
 - a. What mass of lead is in the alloy?
 - b. If the alloy is heated to 350°C (660°F), how much more lead may be dissolved in the α phase without exceeding the solubility limit of this phase?

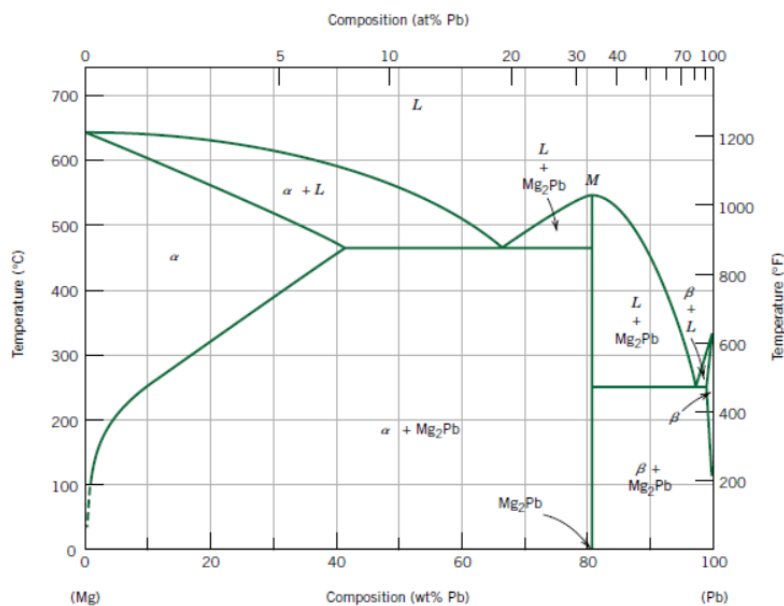


Figure 9.20 The magnesium–lead phase diagram. [Adapted from *Phase Diagrams of Binary Magnesium Alloys*, A. A. Nayeb-Hashemi and J. B. Clark (Editors), 1988. Reprinted by permission of ASM International, Materials Park, OH.]

10. It is desirable to produce a copper–nickel alloy that has a minimum noncold-worked tensile strength of 350 MPa (50,750 psi) and a ductility of at least 48%EL. Is such an

alloy possible? If so, what must be its composition? If this is not possible, then explain why.

11. Figure 9.38 shows the pressure–temperature phase diagram for H_2O . Apply the Gibbs phase rule at points A , B , and C ; that is, specify the number of degrees of freedom at each of the points—that is, the number of externally controllable variables that need be specified to completely define the system.

Figure 9.38
Logarithm
pressure-versus-
temperature phase
diagram for H_2O .

